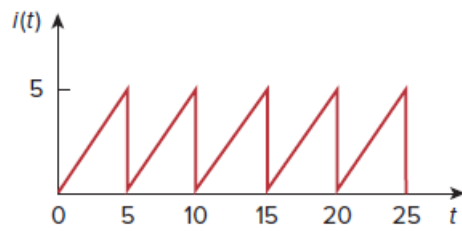


Problem

Calculate the rms value of the current waveform of Fig. 11.58.

Figure 11.58:



Step-by-step solution

Step 1 of 1

$$T = 5, \quad i(t) = t, \quad 0 < t < 5$$

$$I_{\text{rms}}^2 = \frac{1}{5} \int_0^5 t^2 dt = \frac{1}{5} \cdot \frac{t^3}{3} \bigg|_0^5 = \frac{125}{15} = 8.333$$

$$I_{\text{rms}} = 2.887 \text{ A}$$